

A Ten-Paper Fixed-Shift Completion-to-Flatness Audit: Exact Certificate Closures, the Determinant-Energy Gate, and a Zero-Slack Endpoint Ledger

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Abstract

TPC-73–TPC-81 close a sequence of exact primitive-support, direct-entry, convex-dual, provenance, sparse-incidence, complex-scalar forest, anchor–direct, and scalar-spectral interfaces. We audit those nine papers together with the present synthesis as a ten-paper block. The audit separates finite algebra and graph theory (L0), literal fixed-shift transfer (L1), and new asymptotic arithmetic input (L2). Every unconditional theorem in the audited block lies at L0 or L1; none proves a new L2 estimate.

The exact surviving architecture passes through the actual determinant coefficient

$$a_X = (A_{C_X}(n))_n, \quad Z_X = \sum_n A_{C_X}(n), \quad D_X = \sum_n |A_{C_X}(n)|^2.$$

Completion geometry reaches D_X only after a literal determinant-binning map, a positive lower modulus on a verified subspace, and a controlled dictionary residual. It does not control the distinguished zero mode Z_X . At the critical scales

$$Q_X = X^{267/400+o(1)}, \quad J_X = X^{133/400+o(1)},$$

the hypotheses

$$|Z_X| \leq X^{-\eta_Z+o(1)} Q_X^2, \quad D_X \geq X^{-\lambda_D+o(1)} Q_X^3 J_X^2$$

give the flatness bound

$$\frac{|Z_X|^2}{D_X} \leq X^{1/400+\lambda_D-2\eta_Z+o(1)}.$$

Within this two-exponent transfer ledger, the sharp recorded-bounds compatibility condition is $2\eta_Z \geq \lambda_D$. For the current transfer route the present zero-mode target has $\eta_Z = 0$, so the sufficient ledger condition requires natural energy, $\lambda_D = 0$. Support, absolute completion cost, raw energy, anchor quotient geometry, and scalar holonomy cannot supply these two arithmetic inputs without additional literal maps and estimates. We prove no-teleportation and nonidentifiability statements, give branch-specific kill rules, and reduce the next block to three honest routes: determinant-energy survival, distinguished-zero-frequency cancellation, or a negative transfer theorem. These routes concern the TPC-32 matched-shell flatness gate; the exact TPC-18 full-block cover and uniform remainder control remain an independent downstream gate. The paper proves no fixed-shift Chowla estimate, parity breach, prime-pair lower bound, or twin-prime conclusion.

Keywords: fixed shift; proof audit; completion duality; determinant energy; zero-frequency flatness; endpoint ledger; stop certificate.

Convention. Every physical statement retains one fixed nonzero shift h_0 , literal coefficients, complete native keys, exact post-aggregation support, the original quantifier range, and one global raw atomic normalization. No result is specialized to $h_0 = 2$.

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1 Introduction

The purpose of a route audit is not to make a long proof program look closer to its final conjecture than it is. It is to identify which interfaces have become exact, which quantities still require arithmetic information, and which apparent shortcuts are type errors. That distinction is especially important in a fixed-shift problem. An estimate averaged over shifts, frequencies, or carrier families may be deep, yet it does not automatically control one prescribed nonzero h_0 . Likewise, a finite-dimensional completion theorem can be completely rigorous while remaining silent about the signed coefficient to which it is intended to apply.

TPC-73–TPC-80 develop a chain of exact algebraic tools [11–18]. The preceding TPC-72 audit isolated the missing direct two-corner operator bridge and imposed the strict physical loss rule $\Lambda_{\text{phys}} < 1/400$ [10]. TPC-74 closes that direct entry bridge, while TPC-73–TPC-80 close or sharpen a sequence of L0/L1 certificate subgates. Their quantitative aggregation on the small growing physical carrier remains open. TPC-81 then asks for the first time in this block how those tools could reach the determinant coefficient and the distinguished zero frequency used in the matched-shell argument [8, 9, 19]. The answer is both useful and restrictive. Completion can contribute a lower bound for pre-bin energy, but determinant aggregation can erase that energy. Even if determinant energy survives, its unsigned size does not cancel the zero mode. The two tasks must be recorded separately.

This paper is the tenth item in the block. Its main outputs are as follows.

- (i) We give formal definitions of L0, L1, L2 and audit TPC-73–TPC-81 against them. The block contains substantial exact L0 structure and several L1 transfer interfaces, but no completed L2 arithmetic estimate.
- (ii) We write the typed dependency graph. The sparse incidence graph of TPC-77/78, the common-target quotient of TPC-79, the scalar Hermitian row graph of TPC-80, and the determinant coefficient space of TPC-81 are distinct objects. Shared labels do not identify them.
- (iii) We isolate a no-teleportation principle: without an explicit literal synthesis or intertwining map, a theorem on one of those objects supplies no determinant-energy or zero-frequency estimate.
- (iv) We restate the exact anchor–direct and scalar-holonomy interfaces and record their application conditions. In particular, the orphan quotient is not the physical hybrid cost, trees supply no phase-frustration saving, and one sector gap does not give norm strictness.
- (v) We crosswalk the TPC-81 reassembly and determinant-dictionary inequalities into one conditional energy-survival statement. Every factor in that statement is visible; none is silently declared to be $X^{o(1)}$.
- (vi) We prove the recorded-bounds zero-slack compatibility condition

$$2\eta_Z \geq \lambda_D.$$

Equality reaches the TPC-32 flatness threshold but leaves no fixed-power reserve. It is not permission to spend another $1/400$.

- (vii) We give two exact nonidentifiability examples and a three-route decision tree with hard stop rules. These permit a clean negative paper if the two remaining inputs reduce to a fixed-shift Chowla-type correlation, rather than disguising that correlation as a consequence of completion geometry.

The distinction between structural and arithmetic progress is standard in spirit. Sieve parity and multiplicative correlations are not resolved by unsigned support information [2, 4]. Strong averaged correlation theorems provide essential comparison points [5, 6], but they do not become a theorem for the literal distinguished frequency and fixed h_0 without a separate localization argument.

Our convention is deliberately strict. The actual coefficient is formed with all native keys, literal signs and phases, exact post-aggregation support, and one global raw normalization. An ambient support, an unsigned envelope, a fitted reduced space, or a componentwise renormalization may be a diagnostic, but it is not the physical input. This strictness turns several vague obstacles into finite checklists and makes the remaining arithmetic gate precise.

2 Audit levels and the literal packet

Definition 2.1 (The three levels). We use the following labels.

- (1) An L0 result is a finite-dimensional algebraic, convex-dual, combinatorial, graph-theoretic, or spectral theorem. It may be exact and new, but it neither depends on nor automatically applies to the growing arithmetic carrier.
- (2) An L1-*typed conditional interface* has hypotheses that preserve one fixed nonzero h_0 , the complete physical keys, literal coefficients, actual active support, one global raw atomic normalization, and the original quantifier range, and it states how an L0 object connects to the actual growing carrier. Its arithmetic hypotheses may remain open. A *verified physical L1 application* additionally checks every literal packet item and every non-arithmetic interface hypothesis for the carrier under study. Neither label asserts an open asymptotic estimate.
- (3) An L2 result proves a new asymptotic estimate for that literal growing fixed- h_0 carrier, strong enough to reduce a positive exponent in the final physical loss ledger.

Remark 2.2 (Interface progress versus gate closure). An L1-typed interface may identify the right statistic without bounding it. Thus “advances the L1 interface” and “closes the L1 quantitative gate” are different statements. Only the latter can feed an exponent ledger, and it becomes L2 only when the bound is a genuinely new arithmetic estimate.

Definition 2.3 (Literal physical packet). At scale X , a literal physical packet contains:

- (P1) a fixed integer $h_0 \neq 0$, not averaged or selected after the estimate;
- (P2) the complete native key set $\mathcal{K}_X$, including every row, orientation, orbit, divisor, content, source, endpoint, and provenance label used by the construction;
- (P3) the literal coefficient $z_X \in \mathbb{C}^{\mathcal{K}_X}$, with all signs, phases, units, and multiplicities;
- (P4) the exact synthesis maps under discussion, with their genuine domains, targets, and active nonzero support after coefficient aggregation;
- (P5) the determinant labels and every multiplier used in the final binning map;
- (P6) the actual post-aggregation coefficient

$$a_X = (A_{C_X}(n))_{n \in \mathcal{N}_X},$$

with $C_X = \lfloor J_X \rfloor$ in the TPC-32 matched-shell specialization, with cancellations performed only after all native contributions have been assembled;

- (P7) one global raw atomic energy $D_{\text{raw},X} > 0$, computed before deletion, thresholding, block selection, or local renormalization; and
- (P8) the original uniformity and quantifier range needed by the physical conclusion.

Definition 2.4 (Actual support). The actual support of a literal map or coefficient is the set of nonzero entries after exact aggregation with the physical signs and phases. It is not the union of geometrically possible edges, a thresholded numerical graph, or an ambient incidence pattern.

Proposition 2.5 (Promotion test). *An L0 theorem is promoted to a verified physical L1 application only after all items of [definition 2.3](#) used by the theorem have been verified. Omitting any one of fixed h_0 , a native key, actual support, the global raw normalizer, or the required uniformity leaves the result at L0 as a physical application, although a correctly quantified conditional statement may still be L1-typed.*

Proof. Each omitted datum permits two physical packets to have the same reduced model but different target coefficients or normalizations. For example, deleting a native key changes a determinant fiber, replacing actual support by ambient support changes component and spectral data, and local renormalization can hide arbitrary deleted raw mass. Therefore the reduced theorem does not identify the physical statistic. Verification of the full list supplies the needed typing; it does not by itself supply an asymptotic bound. $\square$

Definition 2.6 (Closed interface and closed arithmetic gate). An interface is *closed* when the source, target, map, norm, normalization, and quantifiers are explicit and the asserted identity or inequality is proved. An arithmetic gate is closed only when its required bound is established on the literal packet with the stated asymptotic strength. A finite exact formula for an unknown quantity closes an interface but not necessarily its arithmetic gate.

This terminology is used throughout the table in [section 4](#). It prevents an exact optimization identity from being reported as cancellation, and it prevents a numerical certificate on a truncated carrier from being reported as a uniform asymptotic theorem.

3 The typed dependency graph

The principal completion chain is

$$\begin{aligned}
 & \text{TPC-73: primitive support} \\
 & \quad \downarrow \\
 & \text{TPC-74: direct two-corner entries} \\
 & \quad \downarrow \\
 & \text{TPC-75: completion duality} \longrightarrow \text{TPC-76: provenance lifts} \\
 & \longrightarrow \text{TPC-77: sparse components} \longrightarrow \text{TPC-78: complex-scalar forest and motif reductions.}
 \end{aligned} \tag{1}$$

Two auxiliary branches leave this chain. A verified anchor synthesis and a direct synthesis can enter TPC-79 only if they share one target:

$$\left. \begin{array}{l} \text{direct synthesis into } V \\ \text{anchor synthesis into the same } V \end{array} \right\} \longrightarrow \text{TPC-79: hybrid/orphan quotient.} \tag{2}$$

Likewise, the TPC-80 theorem requires a literal scalar Hermitian matrix H_X^{hol} and a nonnegative majorant P_X^{hol} on the same row graph:

$$(H_X^{\text{hol}}, P_X^{\text{hol}}) \text{ on one scalar actual row graph} \longrightarrow \text{TPC-80: Perron/holonomy audit.} \tag{3}$$

None of (1)–(3) is a determinant-frequency statement. The only typed route from those completion branches to that statistic in the present block is

$$\begin{aligned}
 & \text{literal pre-bin coefficient } \mathcal{B}_X z_X \xrightarrow{\mathcal{J}_X} a_X = (A_{C_X}(n))_n \\
 & \quad \downarrow \\
 & (Z_X, D_X) \longrightarrow \mathfrak{F}_{h_0}(X) \longrightarrow \text{the conditional TPC-32 threshold.}
 \end{aligned} \tag{4}$$

Here $\mathcal{J}_X$ is the exact determinant dictionary, not a label identification.

Theorem 3.1 (Typed dependency theorem). *The implications in (1) are composable only through their stated synthesis and dual maps. The TPC-79 branch additionally requires a common target and separable native variables; the TPC-80 branch additionally requires scalar Hermitian entries and one common actual row graph. A conclusion from either branch enters (4) only through an explicit estimate for the literal $\mathcal{B}_X$, the verified binning map $\mathcal{J}_X$, and its dictionary residual. Shared labels, shared support, or a numerical correlation do not replace these maps.*

Proof. The completion results of TPC-73–TPC-78 concern representations and norms of synthesis targets [11–16]. TPC-79 forms an infimal convolution in one target space [17]. TPC-80 compares quadratic forms of two scalar matrices on one vertex space [18]. By contrast, a_X is indexed by determinant bins. These domains and codomains are not canonically identical. Composition is therefore defined only after the intervening maps and their literal multipliers are specified. TPC-81 makes precisely that extra map explicit [19]. Without it, the displayed objects are merely different typed data sets. $\square$

Remark 3.2 (No silent Perron-pair identification). The TPC-80 pair $(H_X^{\text{hol}}, P_X^{\text{hol}})$ and the TPC-81 normalized missing-Gram pair $(R_{\mathcal{M},X}, P_{\mathcal{M},X})$ have different provenance. TPC-80 can sharpen the missing-Gram estimate only after an exact equality or a norm-preserving intertwining between these pairs is proved on the same actual scalar graph.

Remark 3.3 (Direction of inference). The graph records legal implication, not automatic quantitative success. An exact TPC-78 reduction may return a large optimum; an exact TPC-79 dual witness may certify no saving in the declared hybrid class; and a TPC-80 equality certificate may show exact Perron saturation. These are useful stop certificates but they do not close (4).

4 The TPC-73–TPC-81 stage ledger

Paper	Exact closure	Highest verified stage	Open gate
TPC-73	Canonical primitive support, zero completion, and exact disappearance of every $d \geq 2$ diagonal slice.	L1 local carrier identity.	The actual $d = 1$ mixed variation and a complete operator bound.
TPC-74	Literal normalized two-corner primitive-plane entry for the missing-native Gram, termwise Abel majorants, and a Perron majorant interface.	L1 exact entry representation.	A small $\rho(P_X)$, including the actual row, star, and operator summations.
TPC-75	Strong duality for the completion gauge, quotient seminorms, obstruction witnesses, and an independent row-pair Perron floor.	L0, with L1 typing.	A small optimum on the growing carrier and any genuinely coupled minimax problem.
TPC-76	Divisor-poset zeta lift and inversion, provenance-compatible affine classes, diagonal filtration, and reduced dual problems.	L0/L1 inter-face.	An actual lift whose literal growing optimum gives an asymptotic saving.
TPC-77	Factorization over components of the actual nonzero incidence graph, componentwise primal–dual certificates, and a manifest schema.	L0, advancing L1.	A complete scale-uniform literal census and a quantitative bound on its component optima.
TPC-78	For complex-scalar incidence maps: forest leaf erasure, matching rank, defect reduction, a one-defect closed form, a reduced SOCP, and motif dual packings.	L0.	Proof that the actual carrier is covered by the claimed forests or motifs, with acceptable aggregate cost.
TPC-79	Exact orphan quotient and hybrid duality, together with fixed-datum and global equality tests.	L0, with an L1 admissibility test.	Common physical target, separable variables, correct coefficient prices, and an actual strict saving.
TPC-80	Scalar Hermitian Perron saturation, switching/holonomy classification, sector defects, and weighted defect bounds.	L0, advancing L1 diagnostics.	The actual scalar row graph, quantitative defects in both sectors, and treatment of any block-valued carrier.
TPC-81	Exact completion/reassembly-to-determinant-energy transfer and the zero-mode/energy exponent ledger.	L1-typed conditional interface; no verified physical specialization.	A small dictionary residual, determinant-energy survival, and distinguished-zero-mode cancellation.
Block	Exact support, completion, quotient, sparse, forest, hybrid, scalar spectral, and arithmetic-transfer interfaces.	L0/L1; no L2.	No new fixed- h_0 asymptotic cancellation estimate has been proved.

Theorem 4.1 (Stage audit). *Every unconditional theorem in TPC-73–TPC-81 is an L0 or L1 statement in the sense of definition 2.1. None supplies even the current target*

$$|Z_X| \leq X^{o(1)} Q_X^2, \quad (5)$$

let alone

$$|Z_X| \leq X^{-\eta+o(1)} Q_X^2 \quad \text{for some fixed } \eta > 0, \quad (6)$$

or a polynomially adequate lower bound for the literal determinant energy D_X . Consequently this ten-paper block contains no completed L2 fixed-shift arithmetic estimate.

Proof. TPC-73 and TPC-74 identify literal primitive pieces and entries, but leave their growing operator aggregation open [11, 12]. TPC-75–TPC-80 are exact finite optimization, incidence, reduction, quotient, or scalar spectral theorems; their physical conclusions are conditional on an actual manifest and quantitative hypotheses [13–18]. TPC-81 constructs the exact crosswalk to D_X and Z_X , but states energy survival and zero-mode cancellation as hypotheses [19]. None of these papers proves (5), (6), or the determinant-energy lower bound used in section 8. By definition 2.1, no L2 estimate is present. $\square$

Corollary 4.2 (No arithmetic relabeling). *The labels “exact”, “optimal”, “dual certificate”, and “spectral gap” in the audited block describe their stated L0/L1 objects. They cannot be relabeled as a fixed-shift Möbius saving, a parity-breaking estimate, or a prime-pair theorem.*

Proof. Those arithmetic statements require an L2 estimate for the literal coefficient. Theorem 4.1 shows that such an estimate is not among the premises or conclusions audited here. $\square$

5 A no-teleportation theorem

Four spaces recur in the block:

- (a) the actual nonzero provenance incidence graph $\mathcal{G}_X^{\text{inc}}$ of TPC-77/78;
- (b) the common synthesis target K_X and quotient $K_X/V_{A,X}$ of TPC-79;
- (c) the scalar row space on which the Hermitian pair $(H_X^{\text{hol}}, P_X^{\text{hol}})$ of TPC-80 acts; and
- (d) the determinant-bin space $\mathbb{C}^{\mathcal{N}_X}$ containing a_X .

They can share indices or provenance labels while carrying different operations.

Definition 5.1 (Literal bridge). A literal bridge between two audited objects is a specified linear or affine map, constructed from the complete physical packet, that:

- (B1) preserves all native multiplicities, signs, phases, and normalization factors;
- (B2) has the stated source and target, rather than an observation space with the arrows reversed;
- (B3) maps actual nonzero support to the actual object being estimated; and
- (B4) proves the norm, quadratic-form, or aggregation inequality needed for the next step uniformly in the original quantifier range.

Theorem 5.2 (No teleportation). *Data consisting only of*

$$\mathcal{G}_X^{\text{inc}}, \quad K_X/V_{A,X}, \quad (H_X^{\text{hol}}, P_X^{\text{hol}}),$$

and their support, quotient-gauge, or scalar-spectral invariants do not determine a nontrivial cancellation bound for $|Z_X|$ or a positive lower bound for D_X . Any inference from the listed structural data to such a conclusion requires, at minimum, a literal bridge to the pre-bin coefficient and an exact determinant map

$$\mathcal{J}_X : \mathbb{C}^{\mathcal{Y}_X} \longrightarrow \mathbb{C}^{\mathcal{N}_X}.$$

In particular, the conclusions of TPC-78, TPC-79, and TPC-80 cannot substitute for one another or for a determinant-frequency estimate.

Proof. The listed structural data contain no map to $\mathbb{C}^{\mathcal{N}_X}$. Keeping them fixed therefore imposes no algebraic restriction on a vector assigned to that separate space. One may assign the same structural triple to $a = (1, \dots, 1)$ or, for even dimension, to $b = (1, -1, \dots, 1, -1)$. The two determinant vectors have the same support, entrywise absolute values, and energy, while their zero modes are different. Thus the structural triple does not determine Z_X .

Even a map into a raw atomic space does not determine D_X : a determinant fiber may send two atoms $(1, -1)$ to zero. Hence a positive raw norm does not give a positive post-bin norm without a lower modulus for the actual $\mathcal{J}_X$ on a verified range. These examples are formalized in [section 9](#). A literal bridge and determinant map add exactly the missing relations. $\square$

Corollary 5.3 (Three prohibited identifications). *Without additional proofs, none of the following is valid:*

- (i) treating a four-corner observation or trace as an anchor synthesis vector in the direct target;
- (ii) treating a block-valued carrier as a scalar Hermitian edge matrix, or replacing the actual row graph by an ambient graph; or
- (iii) treating raw atomic mass or pre-bin energy as determinant-bin energy.

Proof. Item (i) reverses or changes the target of a map, item (ii) changes the operator algebra, and item (iii) discards the kernel of $\mathcal{J}_X$. Each violates [definition 5.1](#). $\square$

Remark 5.4 (What the theorem does not say). [Theorem 5.2](#) is a typed nonidentifiability result. It does not assert that no useful bridge exists. Constructing such a bridge is precisely Route I-A of [section 10](#); the direct determinant-energy Route I-B does not require this completion bridge.

6 The TPC-79 and TPC-80 interface audits

6.1 Anchor–direct synthesis in one target

Let K, X_A, X_D be finite-dimensional real or complex spaces, regarded as real when duality is taken. Let

$$S_A : X_A \rightarrow K, \quad S_D : X_D \rightarrow K$$

be real-linear maps, assume S_D is surjective, and let p_A, p_D be the literal coefficient norms. Put

$$V_A = \text{Ran } S_A,$$

$$\alpha(v) = \inf_{S_A x = v} p_A(x), \quad \delta(k) = \min_{S_D y = k} p_D(y),$$

with $\alpha(v) = +\infty$ for $v \notin V_A$. Their polar feasible sets are

$$B_A = \{\lambda : p_A^\circ(S_A^* \lambda) \leq 1\}, \tag{7}$$

$$B_D = \{\lambda : p_D^\circ(S_D^* \lambda) \leq 1\}. \tag{8}$$

Proposition 6.1 (Exact TPC-79 interface). *For the quotient map $\pi : K \rightarrow K/V_A$, define*

$$\delta_{\text{orph}}(\pi k) = \inf_{v \in V_A} \delta(k - v), \tag{9}$$

$$h(k) = \min_{v \in V_A} \{\alpha(v) + \delta(k - v)\}. \tag{10}$$

Then

$$\delta_{\text{orph}}(\pi k) = \max_{\substack{\lambda \in B_D \\ S_A^* \lambda = 0}} [\lambda, k]_{\mathbb{R}}, \tag{11}$$

$$h(k) = \max_{\lambda \in B_A \cap B_D} [\lambda, k]_{\mathbb{R}}, \tag{12}$$

$$\delta_{\text{orph}}(\pi k) \leq h(k) \leq \delta(k). \tag{13}$$

Moreover, with

$$N_D(k) = \{\lambda \in B_D : [\lambda, k]_{\mathbb{R}} = \delta(k)\},$$

one has

$$h(k) = \delta(k) \iff N_D(k) \cap B_A \neq \emptyset, \tag{14}$$

and

$$h \equiv \delta \iff \alpha \geq \delta \text{ on } V_A \iff B_D \subseteq B_A. \tag{15}$$

Proof. These are the quotient-norm and infimal-convolution dual formulas, fixed-datum norming-face test, and global redundancy theorem of TPC-79 [17]. Finite dimensionality gives attainment. $\square$

Proposition 6.2 (TPC-79 admissibility audit). *The formulas in proposition 6.1 apply to a physical anchor/direct proposal only if:*

- (A1) $S_A x$ and $S_D y$ are synthesis vectors in the same target K ;
- (A2) the native variable space is the verified direct sum $X_A \oplus X_D$, or an equivalent separable model is proved;
- (A3) the physical coefficient cost is $p_A(x) + p_D(y)$ with the same global normalization; and
- (A4) both maps use the complete literal packet and actual support.

An observation, trace, or shared label does not imply (A1). Coupled variables or a coupled cost require a new joint polar, not (12).

Proof. The proof of (12) takes the polar of the additive norm on $X_A \oplus X_D$ and uses the common synthesis $S_A x + S_D y$. If the targets, variables, or costs differ, that primal program is not the physical program and its dual intersection is not valid. $\square$

The free orphan quotient δ_{orph} is a lower floor, not the physical hybrid price: it charges zero for every anchor-visible direction. An orphan normer $S_A^* \lambda = 0$ is a strong sufficient certificate for no improvement, but it is not necessary; the exact test is (14). Finally, solver failure proves no obstruction. A no-improvement claim requires an exact normer or dual witness.

6.2 Scalar Perron saturation and holonomy

Let $P \neq 0$ be a real symmetric nonnegative matrix with zero diagonal and connected support graph, and let H be Hermitian on the same scalar vertex space, with zero diagonal and entrywise domination $|H| \leq P$. Write

$$\delta_\sigma = \rho(P) - \lambda_{\max}(\sigma H), \quad \sigma \in \{+1, -1\}.$$

Proposition 6.3 (Exact TPC-80 saturation audit). *One has*

$$\rho(P) - \|H\| = \min\{\delta_+, \delta_-\}. \quad (16)$$

Moreover,

$$\|H\| = \rho(P) \iff |H| = P \quad \text{and} \quad H = \sigma D P D^* \quad (17)$$

for some $\sigma \in \{\pm 1\}$ and diagonal unitary D . When $|H| = P$, writing $H_{ij} = P_{ij} \phi_{ij}$, the switching condition is equivalent to

$$\text{Hol}_\phi(C) = \sigma^{|C|} \quad \text{for every oriented cycle } C. \quad (18)$$

It is enough to test the fundamental cycles of a spanning tree.

Proof. This is the two-sector Perron saturation and scalar holonomy classification of TPC-80 [18]; it also follows from the equality case in the Perron comparison [1, 3]. $\square$

Remark 6.4 (Distinct Perron pairs). In a physical TPC-80 specialization, denote the pair above by $(H_X^{\text{hol}}, P_X^{\text{hol}})$. It is not automatically the TPC-81 missing-Gram pair $(R_{\mathcal{M}, X}, P_{\mathcal{M}, X})$. Using the holonomy theorem inside the missing-Gram certificate requires a proof that $H_X^{\text{hol}} = R_{\mathcal{M}, X}$ and $P_X^{\text{hol}} = P_{\mathcal{M}, X}$ on the same actual scalar graph, or an explicit intertwining that preserves the required quadratic forms and operator norms. Without that bridge, theorem 5.2 applies.

Corollary 6.5 (Tree and two-sector audit). *On a tree, every scalar edge-phase pattern switches away. Hence a tree cannot yield phase-frustration saving. On a graph with cycles, a positive defect in only one sector does not prove $\|H\| < \rho(P)$; both $\delta_+ > 0$ and $\delta_- > 0$ are needed.*

Proof. A tree has no cycle constraint, so (18) is vacuous for either admissible switching sector. The second statement is immediate from (16). $\square$

Proposition 6.6 (Reducible component crosswalk). *For an actual scalar graph with connected components V_j , the domination $|H| \leq P$ gives block decompositions $H = \bigoplus_j H_j$ and $P = \bigoplus_j P_j$, and*

$$\|H\| = \max_j \|H_j\|, \quad \rho(P) = \max_j \rho(P_j).$$

Consequently $\|H\| = \rho(P)$ if and only if some $\rho(P)$ -dominant component satisfies $\|H_j\| = \rho(P_j) = \rho(P)$; on that component the connected saturation criterion of [proposition 6.3](#) applies.

Proof. If i and j lie in different components, then $P_{ij} = 0$, so $|H_{ij}| \leq P_{ij}$ gives $H_{ij} = 0$. The two norm identities follow from block diagonalization. Equality of the two maxima is equivalent to saturation on at least one component attaining the Perron maximum. $\square$

Proposition 6.7 (TPC-80 application boundary). *The scalar theorem does not by itself provide a useful endpoint estimate in any of the following cases:*

- (i) *the physical entries are blocks or operators rather than scalars;*
- (ii) *H and P do not live on the same actual row graph;*
- (iii) *only qualitative strictness $\|H\| < \rho(P)$ is known while $\rho(P)$ itself is too large; or*
- (iv) *no literal bridge relates the row quadratic form to $\mathcal{J}_X \mathcal{B}_X z_X$ or Z_X .*

Proof. Items (i)–(ii) violate the hypotheses of [proposition 6.3](#). In (iii), an arbitrarily small strict gap need not reduce a fixed exponent. In (iv), [theorem 5.2](#) applies. $\square$

7 The TPC-81 completion-to-energy crosswalk

We now record exactly what completion geometry can transfer to the determinant coefficient. Let

$$\mathcal{A}_X, \mathcal{M}_X : \mathbb{C}^{\mathcal{K}_X} \longrightarrow \mathbb{C}^{\mathcal{Y}_X}, \quad \mathcal{B}_X = \mathcal{A}_X + \mathcal{M}_X,$$

be the literal represented, missing, and reassembled maps, and let z_X be the physical coefficient. The terms “represented” and “missing” indicate provenance, not sign: $\mathcal{M}_X$ contains its literal phases.

7.1 The normalized missing Gram

Suppress X temporarily and define

$$G_{\mathcal{M}} = \mathcal{M}^* \mathcal{M}, \quad d_j = (G_{\mathcal{M}})_{jj}.$$

Let the full diagonal operator on $\mathbb{C}^{\mathcal{K}}$ be

$$D_{\mathcal{M}} = \text{diag}(d_j : j \in \mathcal{K}).$$

On the exact active set $K^+ = \{j : d_j > 0\}$, let $G_{\mathcal{M}}^+$ and $D_{\mathcal{M}}^+$ be the corresponding principal restrictions and put

$$R_{\mathcal{M}} = (D_{\mathcal{M}}^+)^{-1/2} (G_{\mathcal{M}}^+ - D_{\mathcal{M}}^+) (D_{\mathcal{M}}^+)^{-1/2}.$$

Let $P_{\mathcal{M}}$ be a verified real symmetric nonnegative majorant satisfying

$$|(R_{\mathcal{M}})_{ij}| \leq (P_{\mathcal{M}})_{ij}$$

on K^+ , and set

$$\hat{\kappa} = 1 + \rho(P_{\mathcal{M}}).$$

If $K^+ = \emptyset$, $P_{\mathcal{M}}$ and $R_{\mathcal{M}}$ are the unique empty matrices and we set $\rho(P_{\mathcal{M}}) = 0$, so $\hat{\kappa} = 1$.

Proposition 7.1 (Coefficient-specific completion survival). *For every z with $\mathcal{A}z \neq 0$, define*

$$\Xi(z) = \hat{\kappa} \frac{\langle D_{\mathcal{M}}z, z \rangle}{\|\mathcal{A}z\|_2^2}. \quad (19)$$

Then

$$\left(1 - \sqrt{\Xi(z)}\right)_+^2 \leq \frac{\|\mathcal{B}z\|_2^2}{\|\mathcal{A}z\|_2^2} \leq \left(1 + \sqrt{\Xi(z)}\right)^2. \quad (20)$$

Proof. The entrywise majorant and Perron comparison give

$$\|\mathcal{M}z\|_2^2 \leq \hat{\kappa} \langle D_{\mathcal{M}}z, z \rangle.$$

Apply the triangle and reverse-triangle inequalities to $\mathcal{B}z = \mathcal{A}z + \mathcal{M}z$, divide by $\|\mathcal{A}z\|_2$, and square. This is the coefficient-specific theorem of TPC-81 [19]. $\square$

Remark 7.2 (Exact but not automatically positive). [Proposition 7.1](#) is an exact transfer inequality. Its lower bound is positive only when $\Xi(z) < 1$. Constructibility of the missing part, or merely finiteness of $\rho(P_{\mathcal{M}})$, does not that condition.

For a common raw normalization, suppose $D_{\text{raw}} > 0$, $\Gamma_{\text{pre}} > 0$, $0 \leq \eta < 1$, and $\chi \geq 0$, and assume

$$\|\mathcal{A}z\|_2^2 \geq (1 - \eta)\Gamma_{\text{pre}}D_{\text{raw}}, \quad (21)$$

$$\langle D_{\mathcal{M}}z, z \rangle \leq \chi\Gamma_{\text{pre}}D_{\text{raw}}. \quad (22)$$

Then TPC-81 gives

$$\|\mathcal{B}z\|_2^2 \geq \left(\sqrt{1 - \eta} - \sqrt{\hat{\kappa}\chi}\right)_+^2 \Gamma_{\text{pre}}D_{\text{raw}}, \quad (23)$$

$$\|\mathcal{B}z\|_2^2 \geq \delta_{\text{reasm}}\Gamma_{\text{pre}}D_{\text{raw}}, \quad (24)$$

$$\delta_{\text{reasm}} := \left[1 - 2\sqrt{\hat{\kappa}\chi} - \hat{\kappa}\chi - \eta\right]_+. \quad (25)$$

The second constant is conservative and ledger-friendly. If its bracket is nonpositive, it is a stop certificate, not a positive survival theorem.

7.2 The determinant dictionary

Let $\nu_X : \mathcal{Y}_X \rightarrow \mathcal{N}_X$ be the literal determinant label and let $\tau_X(y)$ be the archived aggregation multiplier. The exact binning map is

$$(\mathcal{J}_X u)(n) = \sum_{\substack{y \in \mathcal{Y}_X \\ \nu_X(y) = n}} \tau_X(y)u(y). \quad (26)$$

Define

$$r_X = a_X - \mathcal{J}_X \mathcal{B}_X z_X. \quad (27)$$

For a verified subspace E_X containing $\mathcal{B}_X z_X$, let

$$s_X = \inf_{\substack{u \in E_X \\ u \neq 0}} \frac{\|\mathcal{J}_X u\|_2}{\|u\|_2}. \quad (28)$$

If $E_X = \{0\}$, set $s_X = 1$. This is a lower modulus on an independently verified range, not on a one-dimensional span fitted after seeing z_X .

Theorem 7.3 (Literal dictionary-energy transfer). *Assume the complete packet of [definition 2.3](#), the pre-bin data*

$$D_{\text{raw},X} > 0, \quad \Gamma_{\text{pre},X} > 0, \quad \delta_{\text{reasm},X} \geq 0,$$

and the pre-bin bound

$$\|\mathcal{B}_X z_X\|_2^2 \geq \delta_{\text{reasm},X} \Gamma_{\text{pre},X} D_{\text{raw},X},$$

and $\mathcal{B}_X z_X \in E_X$. Then

$$D_X \geq \left(s_X \sqrt{\delta_{\text{reasm},X} \Gamma_{\text{pre},X}} - \frac{\|r_X\|_2}{\sqrt{D_{\text{raw},X}}} \right)_+^2 D_{\text{raw},X}. \quad (29)$$

Proof. By (28),

$$\|\mathcal{J}_X \mathcal{B}_X z_X\|_2 \geq s_X \sqrt{\delta_{\text{reasm},X} \Gamma_{\text{pre},X} D_{\text{raw},X}}.$$

Since $a_X = \mathcal{J}_X \mathcal{B}_X z_X + r_X$, the reverse-triangle inequality and $D_X = \|a_X\|_2^2$ give (29). This is the exact TPC-81 dictionary theorem [19]. $\square$

Corollary 7.4 (Visible energy-loss ledger). *Fix nonnegative exponents $\lambda_{\text{raw}}, \lambda_s, \lambda_\delta, \lambda_\Gamma, \lambda_r$. Suppose*

$$\begin{aligned} D_{\text{raw},X} &\geq X^{-\lambda_{\text{raw}}+o(1)} Q_X^3 J_X^2, \\ s_X &\geq X^{-\lambda_s+o(1)}, \\ \delta_{\text{reasm},X} &\geq X^{-\lambda_\delta+o(1)}, \\ \Gamma_{\text{pre},X} &\geq X^{-\lambda_\Gamma+o(1)} \end{aligned}$$

and, for $0 < \tau_X \leq 1$ satisfying $\tau_X \geq X^{-\lambda_r+o(1)}$,

$$\frac{\|r_X\|_2}{\sqrt{D_{\text{raw},X}}} \leq (1 - \tau_X) s_X \sqrt{\delta_{\text{reasm},X} \Gamma_{\text{pre},X}}.$$

Then

$$D_X \geq X^{-\lambda_D^{\text{cert}}+o(1)} Q_X^3 J_X^2, \quad \lambda_D^{\text{cert}} = \lambda_{\text{raw}} + 2\lambda_s + \lambda_\delta + \lambda_\Gamma + 2\lambda_r. \quad (30)$$

Proof. The positive part in (29) is at least $\tau_X s_X \sqrt{\delta_{\text{reasm},X} \Gamma_{\text{pre},X}}$. Square and collect all losses. $\square$

Remark 7.5 (Status). Proposition 7.1, theorem 7.3, and corollary 7.4 are exact L1-typed conditional interfaces. They do not prove that the bracket in (29) is positive for the actual carrier, that the certified exponent in (30) vanishes, or that the zero mode cancels. Those are the remaining arithmetic inputs.

8 The zero-slack endpoint ledger

For the actual determinant coefficient, define

$$Z_X = \sum_{n \in \mathcal{N}_X} A_{C_X}(n), \quad D_X = \sum_{n \in \mathcal{N}_X} |A_{C_X}(n)|^2, \quad (31)$$

and, when $D_X > 0$,

$$\mathfrak{F}_{h_0}(X) = \frac{|Z_X|^2}{D_X}. \quad (32)$$

If $D_X = 0$, then $a_X = 0$ and hence $Z_X = 0$; the flatness is set to zero by convention. Both statistics in (31) are post-aggregation quantities.

Theorem 8.1 (General two-exponent transfer). *Fix $h_0 \neq 0$ and $C_X = \lfloor J_X \rfloor$. Let*

$$Q_X = X^{\beta+o(1)}, \quad J_X = X^{1-\beta+o(1)}$$

for a fixed $0 < \beta < 1$. Suppose, for fixed $\eta_Z, \lambda_D \geq 0$, that the literal fixed- h_0 coefficient satisfies

$$|Z_X| \leq X^{-\eta_Z+o(1)} Q_X^2, \quad (33)$$

$$D_X \geq X^{-\lambda_D+o(1)} Q_X^3 J_X^2. \quad (34)$$

Then

$$\mathfrak{F}_{h_0}(X) \leq X^{3\beta-2+\lambda_D-2\eta_Z+o(1)}. \quad (35)$$

Proof. The assumed lower bound for D_X is positive for all sufficiently large X . For those X ,

$$\frac{|Z_X|^2}{D_X} \leq X^{\lambda_D-2\eta_Z+o(1)} \frac{Q_X}{J_X^2}.$$

The last ratio is $X^{\beta-2(1-\beta)+o(1)} = X^{3\beta-2+o(1)}$. $\square$

Theorem 8.2 (Zero-slack compatibility). *At the critical scales*

$$Q_X = X^{267/400+o(1)}, \quad J_X = X^{133/400+o(1)}, \quad (36)$$

theorem 8.1 gives

$$\boxed{\mathfrak{F}_{h_0}(X) \leq X^{1/400+\lambda_D-2\eta_Z+o(1)}}. \quad (37)$$

Within this two-exponent transfer ledger, the two recorded bounds imply the TPC-32 condition

$$\mathfrak{F}_{h_0}(X) \leq X^{1/400+o(1)} \quad (38)$$

whenever

$$\boxed{2\eta_Z \geq \lambda_D}. \quad (39)$$

Strict inequality supplies a fixed-power reserve. Equality reaches the threshold and supplies no additional polynomial reserve. The condition is sharp for a deduction from the recorded bounds alone.

Proof. Since

$$3 \cdot \frac{267}{400} - 2 = \frac{1}{400},$$

(37) follows from theorem 8.1. The exponent contributed beyond the baseline $1/400$ is nonpositive exactly under (39). This is a statement about what the two recorded hypotheses imply; it is not a converse characterization of every possible proof of (38). $\square$

Corollary 8.3 (Natural energy in the current transfer ledger). *The current TPC-33 zero-mode target*

$$|Z_X| \leq X^{o(1)} Q_X^2$$

has $\eta_Z = 0$. Since $\lambda_D \geq 0$, the sufficient condition (39) then permits no positive certified loss:

$$\lambda_D = 0, \quad D_X \geq X^{-o(1)} Q_X^3 J_X^2.$$

Thus any positive fixed-power determinant-energy loss cannot close the present transfer from this amplitude bound unless a compensating positive η_Z is proved.

Proof. The sufficient ledger condition becomes $0 \geq \lambda_D$, hence $\lambda_D = 0$. $\square$

8.1 Two ledgers, not two copies of the margin

The $1/400$ in (37) is the geometric baseline Q_X/J_X^2 and the TPC-32 flatness endpoint [8]. It is not a free allowance that can be spent again. Separately, the complete physical route contains activation, represented-frame, missing-mass, native-operator, nuisance-shorting, dictionary, cover, endpoint, and calibration losses. Every fixed-power loss must be derived in the same literal carrier and quantifier scope. This is the strict route rule inherited from the TPC-72 dependency audit [10].

Proposition 8.4 (Strict physical loss rule). *Let Λ_{phys} be the sum of all fixed-power exponent losses actually used in the full physical comparison, after repeated terms have been identified and open terms have been left open. The route may continue only if*

$$\Lambda_{\text{phys}} < \frac{1}{400}. \quad (40)$$

It must stop if

$$\Lambda_{\text{phys}} \geq \frac{1}{400}, \quad (41)$$

including equality.

Proof. The inherited endpoint requires a strict positive remainder after all fixed-power degradations. Equality leaves no such remainder, and an open term cannot be assigned exponent zero. Counting the same baseline or gain twice does not change the literal inequality, so it is excluded from the ledger. $\square$

Remark 8.5 (TPC-81 remains conditional). Theorem 8.2 turns two explicit estimates into the conditional flatness threshold. It proves neither (33) nor (34). In particular, (39) is not itself an L2 input.

9 Two exact nonidentifiability counterexamples

Example 9.1 (Unsigned data do not determine the zero mode). *Let N be even and define vectors in $\mathbb{C}^N$ by*

$$a_n = 1, \quad b_n = (-1)^n, \quad 1 \leq n \leq N.$$

They have identical support, identical entrywise absolute values, and identical energy:

$$\text{supp } a = \text{supp } b, \quad |a_n| = |b_n| = 1, \quad \|a\|_2^2 = \|b\|_2^2 = N.$$

Nevertheless,

$$\left| \sum_{n=1}^N a_n \right| = N, \quad \sum_{n=1}^N b_n = 0.$$

Proposition 9.2 (First nonidentifiability consequence). *Support, absolute coefficient data, total energy, and any completion cost or nonnegative Perron envelope determined solely by those unsigned data do not individually or jointly determine the distinguished zero mode.*

Proof. Every listed unsigned datum is identical for the two vectors in [example 9.1](#), while the zero modes have maximal and zero magnitude, respectively. $\square$

Example 9.3 (Raw atom energy can disappear in one determinant bin). *Let the raw space be $\mathbb{C}^2$, let the determinant space be $\mathbb{C}$, and let*

$$\mathcal{J}(u_1, u_2) = u_1 + u_2.$$

For

$$u^+ = (1, 1), \quad u^- = (1, -1),$$

the raw unsigned data and raw energies agree:

$$|u^+| = |u^-| = (1, 1), \quad \|u^+\|_2^2 = \|u^-\|_2^2 = 2.$$

After binning,

$$\|\mathcal{J}u^+\|_2^2 = 4, \quad \|\mathcal{J}u^-\|_2^2 = 0.$$

Proposition 9.4 (Second nonidentifiability consequence). *A positive raw atomic energy, even together with exact unsigned support and multiplicity, does not imply a positive determinant-bin energy. Such an implication requires a lower modulus for the literal binning map on a verified subspace, or an equivalent range-geometry estimate.*

Proof. [Example 9.3](#) has identical positive raw energy in both cases, while the second vector lies in $\text{Ker } \mathcal{J}$. A lower modulus as in (28) excludes precisely this possibility on the verified range. $\square$

Remark 9.5 (Sharp scope of the examples). These examples do not show that the actual fixed- h_0 coefficient has large zero mode or vanishing determinant energy. They prove that the invariants used before TPC-81 are insufficient to decide those questions. Attaching an example to the literal physical map would upgrade it from an L0 nonidentifiability theorem to an L1 negative diagnostic.

10 The next three routes

The audit reduces the next block to two positive inputs and one negative-transfer alternative:

Route I: determinant-energy survival Route II: distinguished-zero-mode cancellation Route III: exact nonidentifiability or no-go theorem.	(42)
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Routes I and II can proceed in parallel, but their exponents meet only through (39). Route III becomes appropriate when a branch-specific obstruction is proved or both positive routes are shown to require the same unresolved arithmetic input.

10.1 Route I: the literal dictionary and energy survival

The target is

$$D_X \geq X^{-o(1)} Q_X^3 J_X^2, \quad (43)$$

or, after an independent zero-mode gain, the more general inequality $\lambda_D \leq 2\eta_Z$. There are two nonexclusive subroutes.

Route I-A: completion, reassembly, and dictionary. This certificate route has six ordered tasks.

- (I1) Generate a complete literal manifest at every audited scale, retaining the fixed h_0 , all native keys, actual support, literal phases, and the global raw normalization.
- (I2) Construct $\mathcal{J}_X$ from the exact determinant label and archive every multiplier and provenance contribution in each fiber.
- (I3) Run the TPC-77 component census on the actual nonzero incidence graph. Use TPC-78 forest, one-defect, or motif formulas only on components whose literal graph satisfies the corresponding hypotheses.
- (I4) Use TPC-79 only when the anchor and direct variables are separable syntheses into the same target with the physical costs. Otherwise delete the hybrid branch and keep the direct route.
- (I5) Use TPC-80 only on an actual scalar row graph. Audit both sectors, and record exact switching equality rather than interpreting it as cancellation.
- (I6) Prove a positive pre-bin survival factor, a lower modulus s_X , and residual separation strong enough that (29) yields the desired λ_D .

Route I-B: direct determinant-energy analysis. Starting from the already defined literal coefficient $a_X = (A_{C_X}(n))_n$, expand or estimate

$$D_X = \sum_n |A_{C_X}(n)|^2$$

directly. This route does not logically require a completion manifest, a pre-bin carrier, or a lower-modulus argument. It must still preserve the fixed h_0 , the complete definition of $A_{C_X}(n)$, the actual post-aggregation support, and the global normalization. The exact expansion should separate manifestly nonnegative diagonal mass from signed equal-determinant cross-correlations. If a natural-scale lower bound is equivalent to an unresolved fixed-shift Chowla-like correlation, that equivalence is a valid Route-III negative-transfer theorem rather than a proof of energy survival.

Proposition 10.1 (Route-I branch rules). *The following conclusions are local to their branch.*

- (a) *Different targets or coupled variables stop the TPC-79 formula, but do not disprove direct synthesis.*
- (b) *A nonforest component stops the forest shortcut, but leaves the general component SOCP and dual program available.*
- (c) *A tree, or exact switching equality in a dominant scalar component, stops phase-frustration saving on that component, but does not decide amplitude completion or determinant cancellation.*
- (d) *A dual lower bound exceeding the allowed cost disproves only the declared completion class and normalization.*
- (e) *If the currently available Route-I-A certificate yields only $\lambda_D^{\text{cert}} > 0$ while $\eta_Z = 0$, that certificate cannot close the present endpoint transfer. A sharper certificate or a direct Route-I-B estimate remains available. Only a proof that the literal carrier necessarily incurs positive loss is a branch obstruction.*

Proof. Items (a)–(d) follow from the hypotheses and targets of [proposition 6.2](#), [corollary 6.5](#), and [theorem 5.2](#). Item (e) follows from [corollary 8.3](#) together with the branch-locality principle: failure of one sufficient certificate does not prove failure of a sharper lower bound. $\square$

10.2 Route II: the distinguished zero mode

The target is an estimate

$$|Z_X| \leq X^{-\eta_Z + o(1)} Q_X^2, \quad 2\eta_Z \geq \lambda_D, \quad (44)$$

with strict inequality preferred because it supplies reserve. A valid input must concern:

- (Z1) the same prescribed $h_0 \neq 0$;
- (Z2) the actual matched coefficient $A_{C_X}(n)$;
- (Z3) the distinguished frequency $r = 0$, not merely generic $r \neq 0$;
- (Z4) the complete physical block and the original normalization; and
- (Z5) the original uniformity, with no unreturned averaging over shift, frequency, slope, modulus, or carrier family.

Proposition 10.2 (Results that do not close Route II). *None of the following, without a proved localization or transfer back to the literal packet, establishes (44):*

- (i) *an all-but-exceptional-frequency estimate;*
- (ii) *an average over h , slopes, moduli, or frozen families;*
- (iii) *squarefree-support density or an absolute-value Perron bound;*
- (iv) *cancellation on one fiber followed by absolute summation in an outer variable; or*
- (v) *a nonzero-frequency dispersion theorem.*

Proof. Each item changes one of (Z1)–(Z5) or discards the sign interaction defining Z_X . It therefore does not bound the statistic in (44). $\square$

Expanding D_X or Z_X may expose a fixed-shift two-affine Möbius correlation. If the desired bound is shown to be equivalent to a currently unproved Chowla-type estimate, that estimate must be listed as a new arithmetic hypothesis. Averaged Chowla results do not automatically give the required literal form [5, 6].

10.3 Route III: a negative transfer theorem

If Routes I and II both reduce to the same unresolved signed correlation, continuing to rename the input creates no progress. An honest alternative is a paper of the form

Why Completion Geometry Does Not Control the Distinguished Frequency: Nonidentifiability and Stop Theorems for a Fixed-Shift Transfer Architecture.

Its rigorous content can include:

- (N1) identical support, absolute coefficients, and energy with different zero modes;
- (N2) complete loss of raw energy under determinant binning;
- (N3) failure of anchor–direct composition without a common target;
- (N4) absence of scalar phase saving on trees or saturated sectors;
- (N5) failure of scalar Perron strictness to imply determinant flatness; and
- (N6) exact branch-specific dual obstructions for independent, provenance, hybrid, or holonomy models.

Such a result is an L0 no-go theorem. It becomes an L1 negative diagnostic only when its countermodel or dual witness is attached to the literal fixed- h_0 interface. It is not a proof of the general sieve parity barrier and does not disprove a twin-prime strategy outside the declared architecture.

10.4 Operational decision tree

choose a determinant-energy branch:

$$\begin{cases} \text{I-A:} & \text{build the literal manifest and } \mathcal{J}_X, \text{ then estimate } s_X, \delta_{\text{reasm}}, \Gamma_{\text{pre}}, r_X; \\ \text{I-B:} & \text{expand the literal } D_X \text{ directly and isolate its signed cross-correlations.} \end{cases}$$

a physical determinant-energy exponent λ_D known?

$$\begin{cases} \text{no:} & \text{continue Route I or identify the signed correlation obstruction;} \\ \text{yes:} & \text{attack the same coefficient's } Z_X \text{ in Route II.} \end{cases} \quad (45)$$

compare the two exponents:

$$\begin{cases} 2\eta_Z > \lambda_D : & \text{positive reserve; audit the full physical loss ledger;} \\ 2\eta_Z = \lambda_D : & \text{flatness threshold only; no extra loss inside this flatness transfer;} \\ 2\eta_Z < \lambda_D : & \text{stop the current TPC-32 endpoint route.} \end{cases}$$

Even the first case in the last comparison reaches only the conditional matched-shell theorem. A full-block conclusion still requires the exact cover and remainder control isolated in TPC-18 [7].

11 Unified hard stop rules

The following rules apply before any claim is promoted.

- S1.** If the shift is averaged, optimized, or replaced by a moving h , stop the fixed- h_0 claim.
- S2.** If a native key is deleted, actual support is replaced by ambient support, or the raw normalizer is recomputed locally, stop the L1 transfer.
- S3.** If an observation, trace, or four-corner scalar is treated as a synthesis vector without a proved section into the same target, stop the TPC-79 combination.
- S4.** If a block or operator-valued carrier is treated as a scalar Hermitian graph, stop the TPC-80 application.
- S5.** If a numerical solver fails to improve a value but supplies no verified dual witness or exact normer, do not call the outcome an obstruction.
- S6.** When transferring raw or pre-bin energy, if it has not passed through the literal $\mathcal{J}_X$ and a verified lower modulus or residual estimate, do not call it determinant energy. This does not prohibit a direct arithmetic expansion or estimate of D_X .
- S7.** If a dispersion estimate excludes or averages the distinguished frequency, do not call it zero-frequency control.
- S8.** If $2\eta_Z < \lambda_D$, stop the present TPC-32 endpoint transfer.
- S9.** If the complete physical fixed-power $loss$ is $\geq 1/400$, stop; equality also stops.
- S10.** If the TPC-18 full-block cover and its uniform remainder bounds are unverified, do not infer a fixed-shift Hardy–Littlewood, prime-pair, or twin-prime conclusion.

Proposition 11.1 (Scope of a stop certificate). *A stop rule invalidates only the claim or branch named in that rule. It does not prove that every representation, every analytic method, or the underlying number-theoretic conjecture is impossible.*

Proof. Each rule identifies a missing hypothesis of a specified implication. Failure of that hypothesis blocks the implication, not all possible implications with different hypotheses. $\square$

Remark 11.2 (Equality conventions). There are two equalities with different logical roles. Equality $2\eta_Z = \lambda_D$ reaches the conditional flatness threshold but has zero fixed-power robustness. Equality $\Lambda_{\text{phys}} = 1/400$ fails the strict full physical loss rule and therefore stops. Neither equality may be described as spare margin.

Remark 11.3 (Evidence archive). A future literal census should archive the native manifest, the post-aggregation nonzero graph, every connected component, the primal and dual certificate residuals, the scalarization proof where used, all determinant fibers, the lower-modulus subspace, and the global raw normalizer. Numerical evidence is useful for choosing a branch, but promotion requires exact or interval-certified bounds uniform in the stated asymptotic range.

12 Final claim boundary

Theorem 12.1 (Ten-paper completion-to-flatness audit). *The joint unconditional content of TPC-73–TPC-81 and the present paper is the following.*

- (i) *Exact L0 certificate theories are available for primitive support, direct completion, provenance lifts, sparse components, complex-scalar forest/motif reductions, common-target anchor–direct duality for separable variables with additive physical cost, and scalar Perron/holonomy defects.*
- (ii) *Exact L1-typed interfaces identify the conditions under which those theories can be attached to a literal fixed- h_0 carrier; a verified physical application additionally requires the literal packet checks of [proposition 2.5](#).*
- (iii) *TPC-81 gives an exact conditional passage from pre-bin completion survival to determinant energy, with every loss and the dictionary residual displayed.*
- (iv) *The current TPC-32 matched-shell transfer ledger has two explicit arithmetic inputs: determinant-energy survival with exponent λ_D and distinguished-zero-mode cancellation with exponent η_Z . Their transfer ledger closes only under*

$$2\eta_Z \geq \lambda_D.$$

- (v) *Neither input is established by the audited block. Therefore the block proves no L2 fixed-shift arithmetic estimate.*

Proof. Items (i)–(ii) are [theorems 3.1](#) and [4.1](#). Item (iii) is [theorem 7.3](#). Item (iv) is [theorem 8.2](#), and item (v) follows from [remark 7.5](#). $\square$

Corollary 12.2 (What remains genuinely arithmetic). *At the present critical scales, the unresolved positive core of the TPC-32 matched-shell flatness gate is*

$$\boxed{\text{determinant-energy survival} + \text{distinguished-zero-mode cancellation,} \quad 2\eta_Z \geq \lambda_D.} \quad (46)$$

Completion, quotient, and scalar-spectral geometry can quantify interfaces entering the first term, but they do not by themselves prove either term. This is not the whole prime-pair route: even closure of this gate leaves the independent TPC-18 exact full-block cover and uniform remainder control open.

Proof. The determinant-energy crosswalk is [\(29\)](#); zero-mode nonidentifiability is [proposition 9.2](#); and the endpoint comparison is [\(39\)](#). $\square$

The advance of this audit is therefore not a claimed approach to a proved prime-pair theorem. It is a reduction of a diffuse collection of completion heuristics to two typed, quantitative arithmetic questions at the matched-shell gate, together with exact branch-specific certificates for when the algebraic mechanisms cannot answer them. The downstream full-block gate remains separate.

Explicit exclusions. This paper does not prove determinant-energy survival, distinguished-zero-frequency cancellation, a fixed-shift Chowla estimate, a breach of sieve parity, a fixed-shift Hardy–Littlewood asymptotic, a prime-pair lower bound, or the twin-prime conjecture. It does not specialize its conclusions to $h_0 = 2$. The parity problem remains a genuine arithmetic boundary, not a conclusion of the present finite-dimensional analysis [\[2, 4\]](#).

Operational conclusion. Route I is worth continuing while either the literal dictionary or a direct determinant-energy analysis can preserve natural-scale energy, or while Route II supplies enough compensating zero-mode gain. Route II counts only at the actual distinguished frequency and fixed h_0 . If both routes are shown to require the same unresolved Chowla-like correlation, Route III is a mathematically complete negative-transfer outcome. In every case, the complete fixed-power loss remains subject to the strict rule

$$\Lambda_{\text{phys}} < \frac{1}{400}; \quad \Lambda_{\text{phys}} \geq \frac{1}{400} \implies \text{stop},$$

with equality included in the stopping case.

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